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Machine Learning-Based Anomaly Detection in District Heating Buildings Using Substation Data

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Abstract

This thesis investigates the Machine Learning (ML) methods for District Heating System (DHS) anomaly detection using data collected from building substations. Nowadays, DHSes are widely deployed in Northern European countries such as Sweden and play a critical role in providing space heating for buildings. However, operational anomalies in DHS can degrade heating performance and potentially lead to system faults. Therefore, developing a reliable Anomaly Detection System (ADS) is of great importance for maintaining DHS efficiency and reliability.

Given the limited availability of labeled data in real-world applications, this study employs unsupervised ML techniques for anomaly detection, distinguishing it from the more commonly used supervised approaches. The proposed ADS is applied to substations from three distinct DHS-powered building types. It integrates an Auto-Encoder (AE)-based detection algorithm with clustering methods, including Self-Organized Map (SOM) and K-means.

The AE models are used to identify deviations from normal operational behavior, while the clustering algorithms categorize the detected anomalies. Experimental results demonstrate that the proposed models achieve effective anomaly detection across the selected building types. Moreover, the system can also identify not only severe anomalies but also frequent low-level irregularities, thereby supporting both fault prevention and long-term performance optimization of DHS operations.

Keywords

Machine Learning, Anomaly Detection, District Heating System, Substation data, Auto-Encoder

Sammanfattning

Denna avhandling undersöker maskininlärningsmetoder för att upptäcka avvikelser i fjärrvärmesystem med hjälp av data insamlade från byggnaders undercentraler. Numera är fjärrvärmesystem vanligt förekommande i norra europeiska länder som Sverige och spelar en avgörande roll i att tillhandahålla uppvärmning till byggnader. Driftavvikelser i fjärrvärmesystem kan dock försämra värmeeffektiviteten och potentiellt leda till systemfel. Det är därför av stor vikt att utveckla ett tillförlitligt system för avvikelsedetektering för att upprätthålla effektiviteten och tillförlitligheten i fjärrvärmesystemet.

Med tanke på den begränsade tillgången till märkta data i verkliga tillämpningar använder denna studie icke-superviserade maskininlärningstekniker för avvikelsedetektering, vilket skiljer sig från de mer vanligt förekommande övervakade metoderna. Det föreslagna systemet för avvikelsedetektering tillämpas på undercentraler i tre olika typer av byggnader som drivs med fjärrvärme. Det integrerar en algoritm baserad på autoenkoder för att identifiera avvikelser tillsammans med klustringsmetoder, inklusive självorganiserande karta och K-means.

Autoenkodermodellerna används för att identifiera avvikelser från normalt driftbeteende, medan klustringsalgoritmerna kategoriserar de upptäckta avvikelserna. Experimentella resultat visar att de föreslagna modellerna effektivt upptäcker avvikelser i de valda byggnadstyperna. Dessutom kan systemet identifiera inte bara allvarliga avvikelser utan även frekventa låggradiga oregelbundenheter, vilket därmed stödjer både felprevention och långsiktig optimering av fjärrvärmesystemets prestanda.

Nyckelord

Maskininlärning, Avvikelsedetektering, Fjärrvärmesystem, Transformatorstationsdata, Autokodare

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Stockholm, May 2025
Tianze Yao

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List of acronyms and abbreviations

ADS	Anomaly Detection System
AE	Auto-Encoder
ANN	Artificial Neural Network
API	Application Programming Interface
ARIMA	Auto-Regressive Integrated Moving Average
BMU	Best Matching Unit
CDF	Cumulative Distribution Function
CNN	Convolution Neural Network
DH	District Heating
DHN	District Heating Network
DHS	District Heating System
DL	Deep Learning
ED	Euclidean Distance
FN	False Negative
FP	False Positive
GAN	Generative Adversarial Network
HVAC	Heating, Ventilation and Air Conditioning
IoT	Internet of Things
IQR	Inter-Quartile Range
KNN	K-Nearest Neighbor
KTH	KTH Royal Institute of Technology
MAE	Mean Absolute Error
ML	Machine Learning
MSE	Mean Squared Error
PCA	Principle Component Analysis

PDF	Probability Density Function
PID	Proportion Integration Differentiation
RF	Random Forest
SAE	Stacked Auto-Encoder
SDG	Sustainable Development Goal
SOM	Self-Organized Map
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
WCSS	Within-Cluster Sum of Squares

Chapter 1

Introduction

In this thesis, unsupervised Machine Learning (ML) methods are developed and validated to detect operation anomalies in District Heating System (DHS)-powered buildings with three substation data types: Water Flow, Supply Water Temperature and Return Water Temperature. To achieve detection goals, Auto-Encoders (AEs) models are built using substation data from each studied building, and the anomalies are derived with the Mean Absolute Error (MAE) loss function as well as the clustering methods, including K-means and Self-Organized Map (SOM). Finally, anomaly in two focuses are identified with the models, and they are interpreted to the specific anomaly manually based on their behaviors after the detection.

1.1 Background

To ensure a warm indoor environment in winter, buildings are widely constructed with DHS today, *e.g.*, in Sweden the coverage of DHS powered buildings is over 93% [1]. The DHS plays a vital role in providing heating energy to the building and keeping houses or rooms in a warm temperature. Therefore, the reliability and safety of the DHS are essential to the heating performance in buildings. In this thesis, substations in three different types of buildings are measured with intelligent meters and the substation data tracks the supply, return water temperature as well as the water flow in the DHS of each building.

As shown in Figure 1.1, the substation data flow is fed to the Anomaly Detection System (ADS) which contains multiple models. In this thesis, they are an AE system and a loss function system. The AE system in charge of the ideal prediction while the loss function system in charge of the comparison and output anomaly flags as in type of boolean.

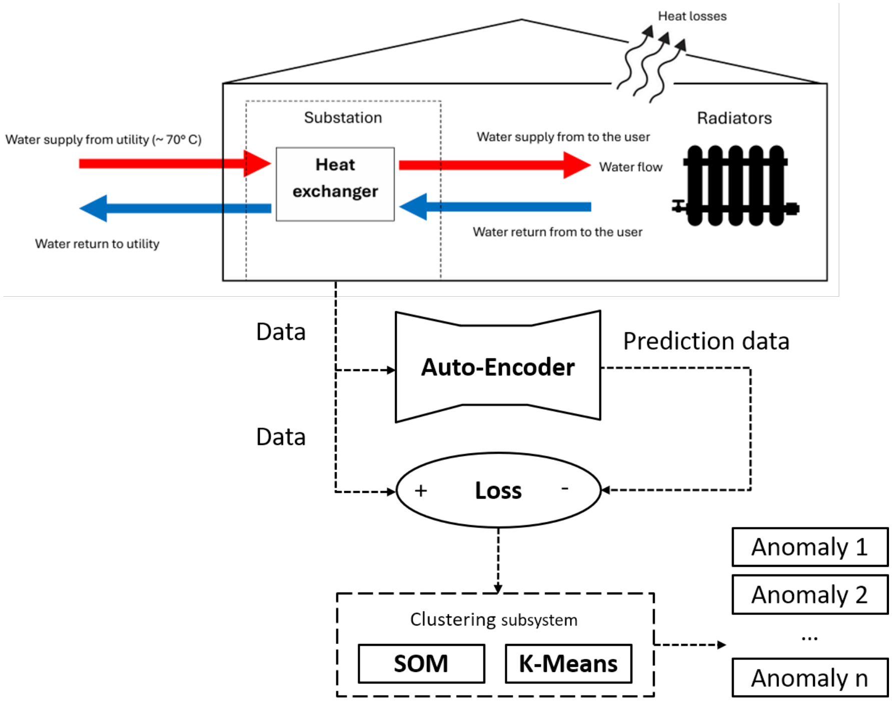

Figure 1.1: ADS flowchart

1.2 Problem

The core research problem in this thesis is how to optimally exploit real-time operation data to precisely detect and identify the anomaly using DHS building substations data.

1.2.1 Original problem and definition

To solve the core problem, several research problems are defined with the step of research, this involves:

- How to leverage ML for abnormal performance detection in DHS-powered buildings using DHS substation data?
- How to utilize the outcome of the proposed ADS to identify the operation anomaly type?

- After detecting operational deviations, how can human expertise be leveraged to determine the cause of each anomaly?

1.2.2 Scientific and engineering issues

The scientific and engineering issues of this thesis work are about the data quality and resource consumption, this includes:

- The low-quality data resulting from the Internet of Things (IoT)-enabled sensors malfunction.
- ML models training is expensive in terms of computing resources.

1.3 Purpose

This research aims to develop a relatively simple yet effective model for anomaly detection that achieves high precision while minimizing computational overhead, thereby improving operational efficiency. The findings are particularly relevant to building managers and researchers in the field of building energy systems, as the proposed approach not only accurately detects anomalies—including minor deviations—but also identifies potential issues in energy delivery and DHS operations.

1.4 Goals

The goal of this project is having a reliable and systematic ML method for DHS anomaly detection. This has been divided into the following three sub-goals:

- Build a precise and interpretable model for DHS anomaly detection using three data DHS metrics.
- Successfully capture operation deviation and anomalies in different short and long-term time periods.
- Generalize the study to different building categories, *e.g.*, office, public, commercial, *etc*.

The results will contain a reliable model for anomaly prediction, and several case studies as well as the detection results in real buildings and their DHSes.

1.5 Research Methodology

The methods for the research are divided into three parts:

1.5.1 Data Preprocessing

The data used in this study consists of time-series measurements collected over several years from a District Heating (DH)-powered community in Großschönau, Austria. These data are recorded by smart meters, transferred using IoT communication and stored on a local database at KTH Royal Institute of Technology (KTH). To retrieve the data for research purposes, an Application Programming Interface (API) is utilized. By using the designated access link, data corresponding to a specific time range can be extracted in JSON format.

However, due to frequent short-term shutdowns of the smart meters, the raw dataset contains numerous missing values, rendering it unsuitable for direct use in model training. Given the brief duration of these data gaps, a reasonable preprocessing strategy is to skip the missing entries and reshape the remaining data into a format compatible with the ML models.

1.5.2 Model Construction

Given the nature of substation data, this research focuses on time-series prediction for anomaly detection. A variety of models are available for time-series forecasting, including Artificial Neural Network (ANN), Auto-Regressive Integrated Moving Average (ARIMA), and AE, *etc.* [2]. To identify anomalies, a threshold must be established to evaluate the model's predictions. This threshold is typically determined by comparing predicted values with historical "healthy" data. However, in this thesis, the dataset is unlabeled and lacks any predefined health indicators. Therefore, the model is designed to learn the most frequent or typical patterns within the time series and to flag deviations as anomalies, using statistical criteria such as the 3σ (Three-sigma) rule.

The substation data used in this study were collected from real-world buildings and are characterized by a significant amount of noise. For model selection, the review conducted by Berahmand *et al.* [3] highlights the strong performance of AEs in anomaly detection tasks. AE models are shown to be applicable across a wide range of domains, offering interpretable structures and low computational demands. As summarized in Table 1.1, AE is compared with several alternative models, including ANN, Convolution Neural Network (CNN), and Generative Adversarial Network (GAN). While AE achieves comparable performance to these models, it requires fewer computational resources. For this reason, AE is selected as the unsupervised ML model for anomaly detection in this study.

Table 1.1: Model comparison [4]

	AE	ANN	CNN	GAN
Learning style	Unsupervised	Supervised	Supervised	Unsupervised
Supported data	Any	Any	Figure and time	Voice, text and figure
Dimension reduction ability	Yes	No	No	No
Feature extraction ability	Strong	Medium	Strong	Strong
Training difficulty	Low to medium	Low	High	High
Explanatory difficulty	Easy	Easy	Medium	High

1.5.3 Result Interpretation

For the detection result, it's predictable that large amount of anomaly flags will rise in a long time period. To facilitate effective interpretation, clustering and classification methods, such as K-means, SOM, K-Nearest Neighbor (KNN) and Random Forest (RF), are essential for identifying the dominant types of anomalies and quantifying their occurrences. In this thesis, K-means and SOM are selected to cluster the anomaly flags due to their low computational requirements, high accuracy, and efficient training processes.

Regarding anomaly interpretation, Neumayer *et al.* [5] suggest that in typical DHS applications with relatively small datasets, manual analysis remains the most accurate approach, despite being less efficient than automated classification. In line with this, the dataset used in this thesis is of limited size, making it suitable for manual analysis. Therefore, anomaly interpretation is conducted manually by comparing the flagged data points with their neighboring time points to identify deviations and assess potential causes.

1.6 Delimitations

The limitations of this thesis primarily relate to restricted data access. Due to permission constraints, substation data could only be collected from 6 buildings. As a result, the dataset is insufficient for conducting large-scale or community-level DHS network analysis, and the scope of investigation is limited to the DHS operations within these selected buildings.

In terms of model development and prediction, a major limitation is the absence of labeled healthy data. This lack of reference data prevents the model from being trained on a fully healthy baseline, potentially reducing the accuracy of anomaly detection. Furthermore, frequent missing values in the dataset negatively affect the continuity of the time series, which may further impair model performance.

Regarding anomaly interpretation, the absence of formal anomaly reports introduces additional challenges. While anomalies can be detected by the model, their classification and interpretation become more subjective, relying heavily on expert judgment. Although manual analysis improves interpretive accuracy, it also increases the time and effort required for analysis due to the lack of standardized reference data for validation and comparison.

1.7 Structure of the thesis

Chapter 2 presents relevant background information about AE, threshold setting methods and the used clustering methods for anomaly flags. Chapter 3 presents the methodology and method used to solve the problem. Chapter **??** and Chapter 4 present the implementation and the results of the thesis work. The last two chapters discuss and conclude the thesis work as well as come up with the future work.

Chapter 2

Background

This chapter provides basic background information about AE model used in the thesis work. Additionally, this chapter describes the threshold setting methods and the clustering methods, which are used to interpret the anomaly. The chapter also describes related work about the anomaly detection in DHS.

2.1 Auto-Encoder

This section demonstrates the necessary knowledge of AE needed for the prediction, including the structure, the algorithms of AE and the design of loss functions.

2.1.1 Structure

The structure of AE is shown in Figure 2.1, it has three core parts: encoder, latent layer (showed as $Code$ in the figure) and decoder. Each layer is constructed with one or multiple neural networks, which aims to extract or expand the features of the layer's input. For the AE that has multiple layers in the encoder and the decoder, it's also known as Stacked Auto-Encoder (SAE), and SAEs is selected as the prediction models in this thesis due to its stronger learning ability than simple AEs.

While AE is doing anomaly detection, it's an unsupervised Deep Learning (DL) model. It's based on the back propagation algorithm and optimization methods (such as gradient descent method), using the input data itself as supervision to guide the neural network to learn a mapping relationship with the input data, so as to obtain a reconstructed output. In the time series anomaly detection scenario, anomalies are rare compared to normal, so it's

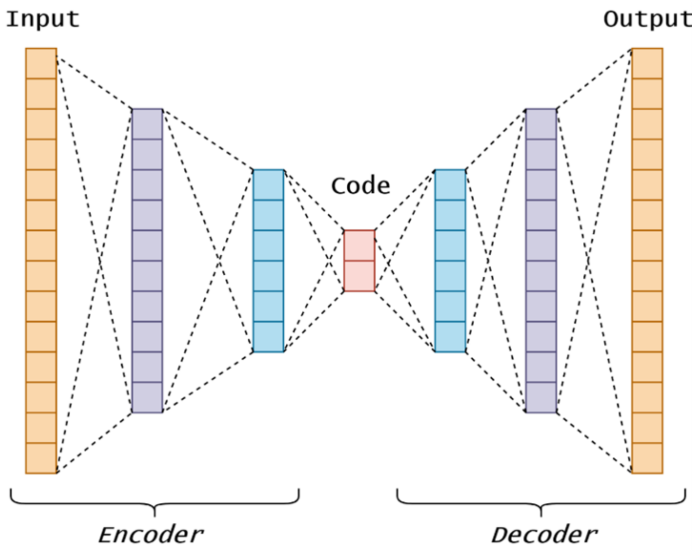

Figure 2.1: The structure of Auto-Encoder [6]

believed that if the difference between the output reconstructed by the AE and the original input exceeds a certain threshold, the original time series has anomalies.

2.1.2 Algorithms

The algorithms of the model consists of two parts: encoder and decoder. The encoder's algorithm is similar to the Principle Component Analysis (PCA), which reduces the dimension of the input so that the key features can be recorded in the latent layer. The decoder receives the latent layer data as input and reconstructs it.

The role of the encoder is to encode the high-dimensional input $\boldsymbol{X}$ into a low-dimensional latent variable $\boldsymbol{h}$, thereby forcing the neural network to learn the most informative features; the role of the decoder is to restore the latent variable $\boldsymbol{h}$ of the hidden layer to the initial dimension $\boldsymbol{X^R}$. The best case is that the output of the decoder can perfectly or approximately restore the original input, *i.e.*, $\boldsymbol{X} = \boldsymbol{X^R}$. The processes of encoder and decoder can be described as Equation (2.1) and Equation (2.2), respectively, where $\boldsymbol{W_1}$, $\boldsymbol{W_2}$ are the initial weights of neurons in two layers, $\boldsymbol{b_1}$, $\boldsymbol{b_2}$ are the bias values of two layers.

$$\boldsymbol{h} = f_\theta(\boldsymbol{X}) = \sigma(\boldsymbol{W_1 X} + \boldsymbol{b_1}) \tag{2.1}$$

$$\boldsymbol{X^R} = g_\phi(\boldsymbol{h}) = \sigma(\boldsymbol{W_2 h} + \boldsymbol{b_2}) \tag{2.2}$$

2.1.3 Loss functions

After the reconstruction process, a loss function is applied to measure the difference between the input and the output. The loss function can be determined as Equation (2.3), where the *dist* function measures the distance (*i.e.*, difference). The loss function can be chosen from Mean Squared Error (MSE) or MAE, *etc*.

$$Loss(\boldsymbol{X}, \boldsymbol{X^R}) = dist(\boldsymbol{X}, \boldsymbol{X^R}) \tag{2.3}$$

Considering the issues of overfitting, the loss function can be designed by adding several components to mitigate or prevent the overfitting by reducing some neurons' activities [3]. The expanded loss function can be written as Equation (2.4).

$$Loss(\boldsymbol{X}, \boldsymbol{X^R}) = dist(\boldsymbol{X}, \boldsymbol{X^R}) + \eta\rho(\boldsymbol{Z}) + \lambda||\boldsymbol{W}||_2^2 \tag{2.4}$$

In general, there are two common ways to solve the overfitting problem. The first way is adding a sparse component (showed as the second component in Equation (2.4)), where $\rho(\boldsymbol{Z})$ determines how many neurons are active in the network and the parameter η determines the intensity of the sparse component. The second way is add regulation component to neurons, which is showed as the last component in Equation (2.4). In this component, $\boldsymbol{W}$ is the regularization function and parameter λ determines the intensity. The commonly used regularization functions include *L1* and *L2* functions, the *L2* function is showed in Equation (2.4) and utilized in this thesis.

2.2 Threshold setting method

In this section, the method used to set threshold for the models is presented. In this thesis work, the methods of 3σ criteria and Inter-Quartile Range (IQR) are considered to set threshold for deviations in three substation data types.

2.2.1 Three-sigma criteria

Three-sigma criteria, also known as three-sigma rule, is a criteria that works for the normal distributed data [7]. This criteria sets the threshold in value ranges for a normal distributed dataset, it describes the distribution by using two statistical variables: mean value and standard deviation.

The normal distribution can be described as

$$X \sim N(\mu, \sigma) \tag{2.5}$$

where μ is the mean value and σ is the standard deviation, and the variable X obeys normal distribution. The Probability Density Function (PDF) of X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{2.6}$$

With the PDF, the Cumulative Distribution Function (CDF) of X is calculated as

$$F(x) = \int_{-\infty}^{x} f(t)dt = \frac{1}{2}[1 + erf(\frac{x-\mu}{\sqrt{2}\sigma})] \tag{2.7}$$

where *erf* is the error function. In 3σ rule, we are interested in the probability of taking k standard deviations to the left and right of μ, that is

$$P(\mu - k\sigma \leq X \leq \mu + k\sigma) = F(\mu + k\sigma) - F(\mu - k\sigma) \tag{2.8}$$

For k = 1, 2, 3, the calculation results are shown in Table 2.1. Proposed by Pukelsheim *et al.* [7], the data outside 2σ can be seen as the significant outliers, and it's almost no data outside the 3σ range. With this knowledge, the boundary between normal and anomaly data can be set and adjusted by different k values. The visualization for the common k values (*i.e.,* 1, 2, 3) is shown in Figure 2.2.

Table 2.1: The included range for different k values

k	1	2	3
Included percentage	68.27%	95.45%	99.73%

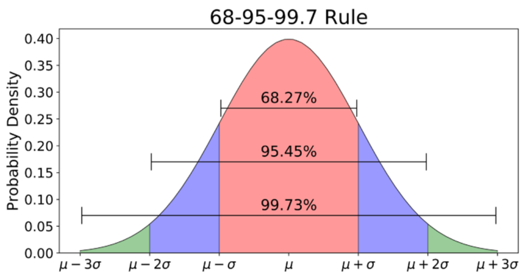

Figure 2.2: 3σ rule visualization [7]

2.2.2 IQR criteria

When the data doesn't obey the normal distribution, the rule for threshold is also different. A common and practical method is IQR criteria, which is applicable to the data without normal distribution. According to the definition of box plot, the outliers are defined as the data who outside the range between up whisker and low whisker. The two whiskers are calculated with the first and the third quartiles of the data.

For a whole dataset, the quartiles are the points who divide the dataset to four equal segments, and the IQR is calculated as

$$IQR = Q3 - Q1 \tag{2.9}$$

where *Q3* and *Q1* are the third and the first quartiles. With IQR, the two whiskers can be defined as

$$Whiskers = Q3 \pm k * IQR \tag{2.10}$$

where the parameter *k* can be set to adjusted the whiskers according to the request, generally it's set to 1.5. The data outside the whiskers is regarded as anomaly, as is showed in Figure 2.3, and for more extreme cases, *k* can be set to 3 or more.

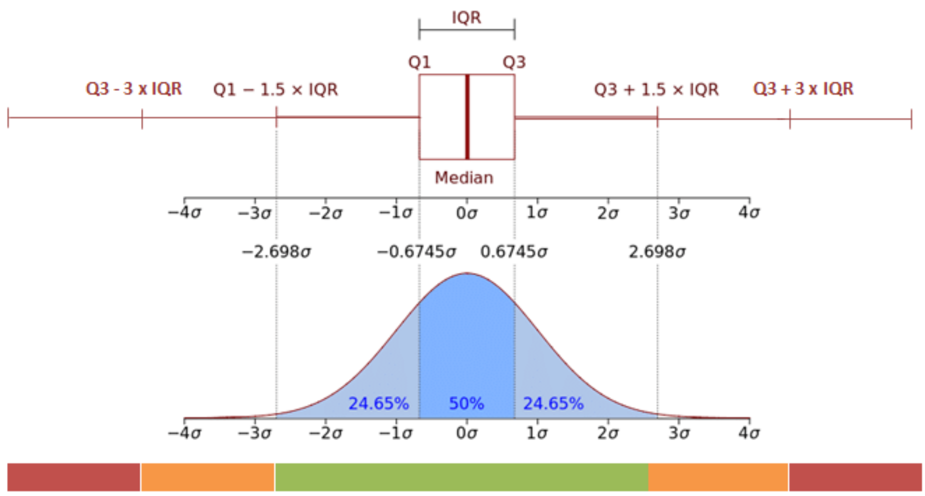

Figure 2.3: IQR visualization [8]

2.2.3 Clustering methods

After detecting the anomaly, it's crucial to classify them for best interpretation. In this subsection, two classical and practical clustering methods are demonstrated and they are used in the thesis work.

2.2.3.1 K-means clustering

K-means clustering is a widely used algorithm based on the iterative allocation and updating of centroids. It is considered one of the most effective clustering methods due to its low computational complexity and high efficiency [9]. The flowchart of the K-means algorithm is presented in Figure 2.4.

The algorithm begins by specifying the desired number of clusters, which determines the initial number of centroids. Subsequently, the Euclidean Distances (EDs) between each data sample and the centroids are calculated, and each sample is assigned to the nearest centroid. If the new assignment differs from the previous iteration, the centroids are updated by computing the mean position of all samples within each cluster. This process is repeated until the cluster assignments stabilize, *i.e.*, no further changes occur in the allocation of samples.

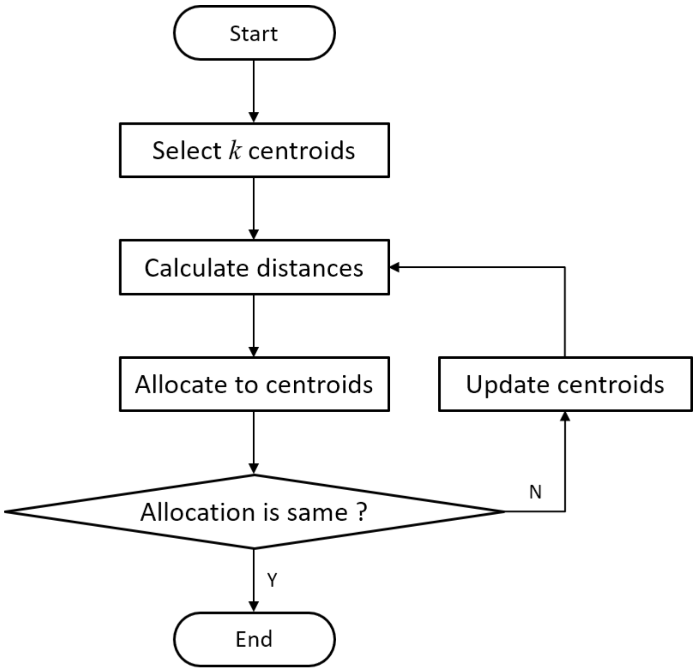

Figure 2.4: K-means flow chart

With this knowledge, the aim of K-means is to minimize the Within-Cluster Sum of Squares (WCSS), *i.e.*, the sum of the squares of the distance from each sample to the centroid of the cluster which it belongs. The aim function is shown in Equation 2.11, where K is the number of clusters, C_i is the sample set of the i^{th} cluster. $||x-\mu_i||^2$ means the ED between the sample x and the centroid μ_i, J represents the aim, *i.e.*, WCSS.

$$J=\sum_{i=1}^{K}\sum_{x\in C_i}||x-\mu_i||^2 \tag{2.11}$$

The update step for centroids can be shown as Equation 2.12. The updating is completed with the position means of each samples in the cluster.

$$\mu_j=\frac{1}{|C_j|}\sum_{x\in C_j}x \tag{2.12}$$

2.2.3.2 SOM clustering

The SOM clustering is a method based on the self-updating of a neural network. The structure of SOM is shown in Figure 2.5. There are two layers in SOM: the input layer and the competition layer. The input layer contains all samples that need to be clustered. When the clustering process begins, samples are sent to the competition layer once a time randomly.

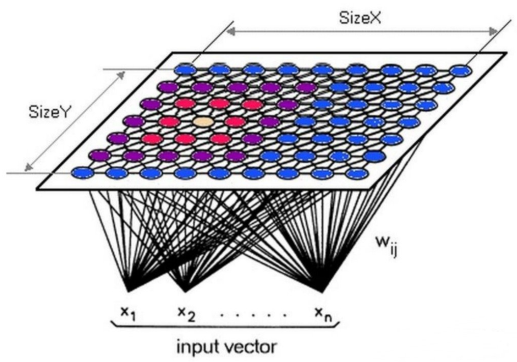

Figure 2.5: SOM structure

Proposed by Kohonen [10], the SOM algorithm can be analyzed from the network structure and the positions of samples. For each input sample, the similarities (usually represented with EDs) with nodes in competition layer are calculated and the most similar node (*i.e.*, the node has least ED) in the competition layer is called Best Matching Unit (BMU). With a specific neighbor range for BMUs, the nodes inside the range are updated with a neighbor function. The nodes who are closer to the BMU will be updated with higher intensity, and the update intensity also decreases with time. For a input sample $\boldsymbol{X_i}$, the update processes for i^{th} BMU and the nodes inside the neighbor range can be described as

$$\boldsymbol{W_i}(t+1) = \boldsymbol{W_i}(t) + \alpha(t) \cdot h_{ci}(t) \cdot (\boldsymbol{X_i} - \boldsymbol{W_i}(t)) \tag{2.13}$$

where $\alpha(t)$ is the learning rate and $h_{ci}(t)$ is the neighbor function, they both decrease with time. For the neighbor function, we always use Gaussian function:

$$h_{ci}(t) = e^{-\frac{||r_c - r_i||^2}{2\sigma(t)^2}} \tag{2.14}$$

where $||r_c - r_i||^2$ calculates the ED between BMU and the i^{th} node in the neighbor range, $\sigma(t)$ is the neighbor range which decreases with time.

With this knowledge, a significant advantage of SOM is that the definition of cluster number is not necessary. The clustering process ends when the max iteration time is reached, or the learning rate and neighbor range are small enough. The clustering result is presented by the final weighs of nodes in the competition layer.

2.3 Related work area

The related work section focuses on three parts of previous work: data preprocessing, anomaly detection and interpretation, which follows the work order of this thesis.

2.3.1 Data preprocessing

In this study, three types of substation data: Water Flow, Supply Water Temperature and Return Water Temperature will be analyzed and examined. To better make training set and testing set with the raw substation data, Van Dreven *et al.* [11] propose preprocessing methods that involve time-series segmentation and random sampling across different seasons.

As showed in Figure 2.6, the proposed in [11] contains not only the segmentation, but also the random sampling from the data and the distance matrices calculation. This approach enables the analysis to focus on specific time periods rather than the entire recording duration, while also enhancing the dataset's representativeness by mitigating the influence of seasonal variations. Particularly, the distance matrices also allows for small time shifts to reflect minor delays in the DH network but prevents overly aggressive realignments that could distort the true relationships.

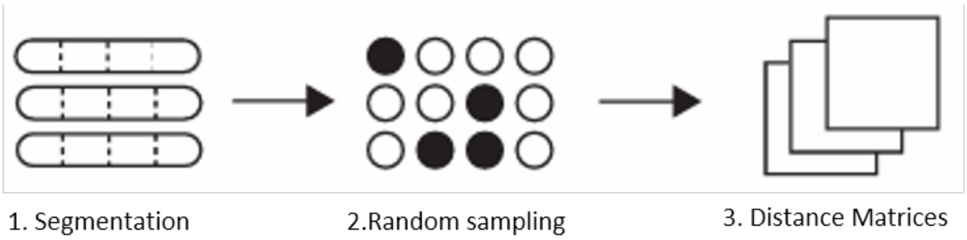

Figure 2.6: A proposed data preprocessing workflow [11]

In addition to addressing the characteristics of different data types, it is essential to ensure that the data are suitable for ML training and prediction. This involves eliminating or skipping missing values, as well as selecting data segments of manageable length. Datasets with a high proportion of missing values are considered unsuitable for analysis [11]. Furthermore, Stecher *et al.* [12] introduce a feature-based clustering approach that facilitates the automatic and efficient construction of training datasets, enhancing both data quality and preprocessing efficiency.

2.3.2 Detection in substation anomaly

Anomaly detection in substation data is performed based on the three key variables previously described. The objective is not only to detect anomalies by raising alerts, but also to identify and classify specific anomaly types. In this study, unsupervised ML methods are employed to flag anomalous behavior, while clustering techniques are used to categorize these anomalies, thereby reducing the manual workload required for interpretation.

For model development, Neumayer *et al.* [5] review a range of anomaly detection strategies in DHS applications, including both threshold-based and clustering-based methods. Due to the absence of reported faults in this research, expert-driven interpretation plays a crucial role. As a result, threshold-based methods are primarily considered. In this context, Losi *et al.* [13] propose a neural network-based approach that establishes thresholds using labeled healthy datasets.

However, since no healthy reference data are available in this study, alternative strategies must be adopted. Sun *et al.* [14] introduce clustering-based methods capable of detecting anomalies without relying on healthy data, while Hermans *et al.* [15] present techniques for threshold setting using statistical approaches such as the three-sigma rule and regression models. To enhance detection efficiency, Mavikumbure *et al.* [16] demonstrate the strong performance of AE-based methods for anomaly detection in building systems, making them a suitable and promising choice for the present research.

2.3.3 Anomaly identification and interpretation

Once anomaly flags are generated by the model, accurate interpretation and identification of the anomalies become essential. Leiria *et al.* [17] propose a clustering approach based on anomaly characteristics and further identify potential issues such as anomalies in Return Water Temperature. Similarly, Sun *et al.* [14] provide a detailed quantification framework for common anomaly types, including leakage, sensor malfunction (*e.g.*, damaged temperature probes), and inaccurate heat meter readings.

Building upon these methods, the interpretation process in this research relies heavily on manual analysis, given the absence of labeled anomaly data. Consequently, the comparison between anomalous and normal operational data plays a critical role in validating and understanding the detected anomalies. With the knowledge of [14], some possible anomalies and its manual analysis standards are shown in Table 2.2 but they are not bound to exist in this thesis work, the standards are only for reference.

Table 2.2: Features of some anomalies [14]

Anomaly interpretation	Instant flow/$m^3 \cdot h^{-1}$	Differential temperature/$^{\circ}C$	Heat power/kW
Leakage, opening windows, extra heat exchanger	>0.8 or normal	>17	>6.2
Inaccurate heat meters	$0 \sim 1$	<0	0
Temperature probe damage	<0.1 or normal	>20	<3.38

2.4 Summary

In the background section, various methods regarding data preprocessing, anomaly detection and interpretation are presented. In this thesis work, they are used in different aspects of the thesis work. However, they may be chosen with the consideration of their advantages and shortcomings during the work.

Chapter 3

Methods

The purpose of this chapter is to provide an overview of the research methods used in this thesis. Section 3.1 describes the research process. Section 3.2 focuses on the data collection techniques used for this research. Section 3.3 describes the experimental design. Finally, Section 3.4 describes the framework selected to evaluate the final interpretation results.

3.1 Research Process

Figure 3.1 shows the steps for the anomaly detection. In this research process, the input layer contains the preprocessed substation data. With specific weight $\boldsymbol{W}$ and bias $\boldsymbol{b}$, the input data is trained and predicted by the AE (z in Figure 3.1). The output data will be compared with the input data as an ideal reference, the comparison relies on MAE loss function. The threshold for the normal distributed loss is set via three-sigma criteria and the performance deviation who exceeds the threshold is marked with an anomaly flag.

Figure 3.2 shows the clustering workflow for anomaly. To analyze the loss values—measured by MAE—a SOM is employed to project the data into a two-dimensional heatmap. Each data sample is mapped to a BMU on the SOM grid. Based on this visualization, the BMUs are further clustered using the K-means algorithm to identify representative groups of common anomalies, represented by the K-means centroids.

These centroids serve as prototypes of typical anomaly patterns and can be manually interpreted to determine their corresponding physical or operational meanings. Subsequently, all samples assigned to the same cluster can be classified as exhibiting the same type of anomaly, as represented by the respective centroid.

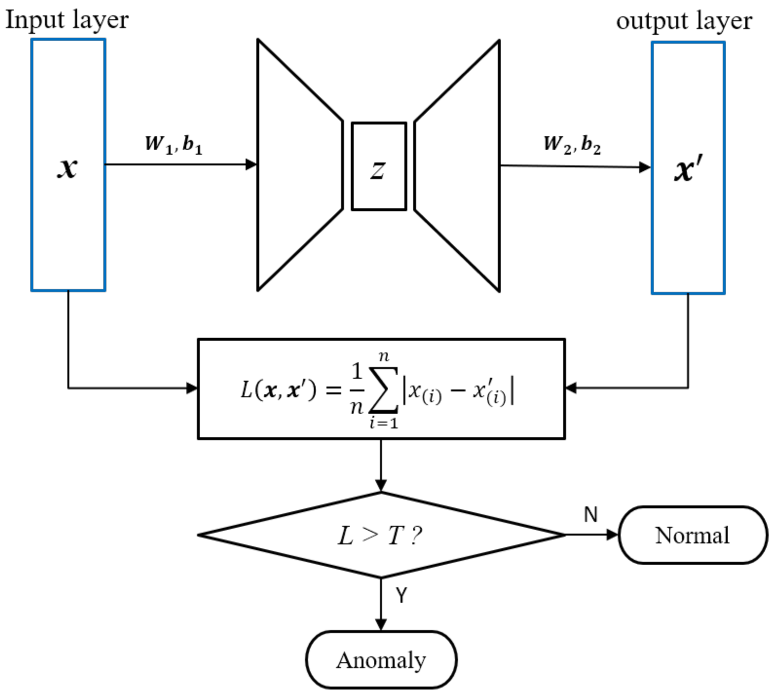

Figure 3.1: Anomaly detection workflow

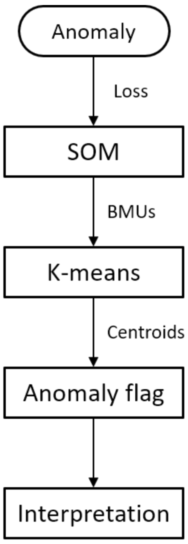

Figure 3.2: Clustering workflow

3.2 Data collection and preprocessing

The data collection is done with the API methods. In a remote database, the substation data of six buildings in three types is allowed to retrieve, the download format is JSON. All data collected from the database is legally permitted by the owner, and they are only used for this thesis work, the relevant rules about data protection are strictly obeyed.

Besides the collection methods, the preprocessing methods are also presented in this section. The preprocessing aims to mitigate the negative effect caused by missing values as well as create a high quality dataset for ML models.

3.2.1 Substation data source

As shown in Figure 3.3, the substation data is collected from a complete district heating network in Großschönau community, Austria. The types of buildings are various from public (*e.g.*, hotel and commercial buildings) to private (*e.g.*, private residence), but in this study the residential buildings are excluded due to data privacy issue. The blocks in Figure 3.3 mark the buildings, the red and blue lines mark the supply and return water pipes, respectively. Each building in this community is equipped with a substation to regulate the water for heating, and the pipes are connected from the master substation to all buildings.

3.2.2 Sampling

The data is recorded starting from July 2023 with the sample rate of 1 point/min. The data types are Supply Water Temperature ($°C$), Return Water Temperature ($°C$) and Water Flow (*kg/s*) recorded in the secondary side of the DHS heat exchanger of each heat consumer.

The sample size is defined as one week based on the data sampling rate. The substation data used in this study covers the period from January 2, 2024, to January 29, 2024, excluding 10 days during which the meters were shut down for maintenance. Under these conditions, the total number of data points amounts to approximately 21,200 per variable per building.

Due to the presence of missing values within the dataset, the effective sample size is set to 4,800 data points. This size is derived from the total available data after accounting for missing segments, and it is used consistently as both the input and output dimension for the AE model.

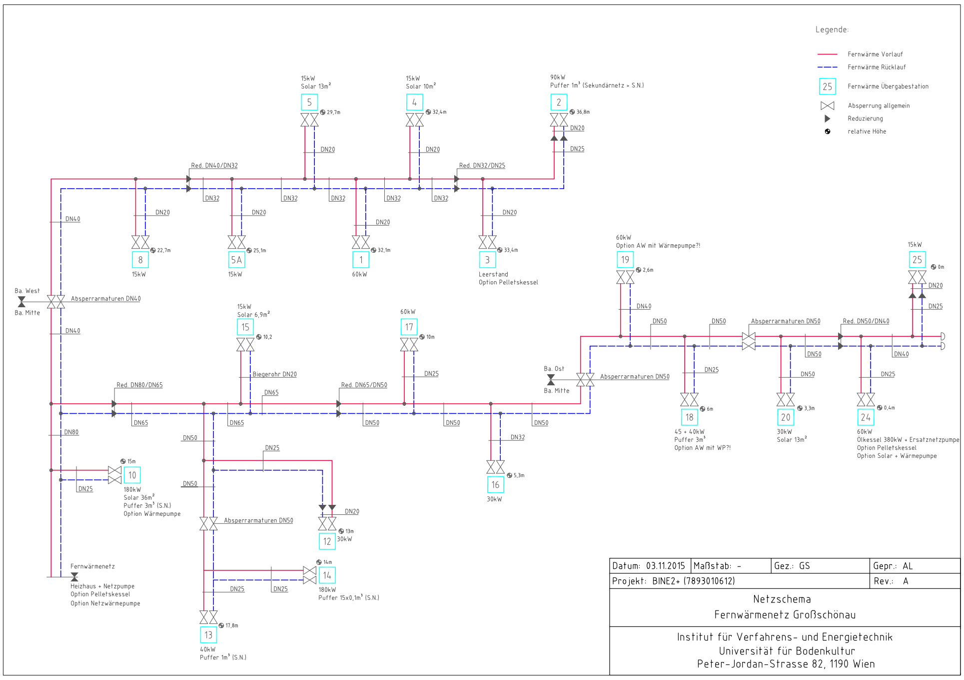

Figure 3.3: DHS network of the Großschönau community

3.2.3 Dataset creating

For the collected substation data, the training set is made without missing values by skips to ensure the continuity. In this thesis work, the substation data has the calculated missing rate of 18.2%, which is suitable for the model training.

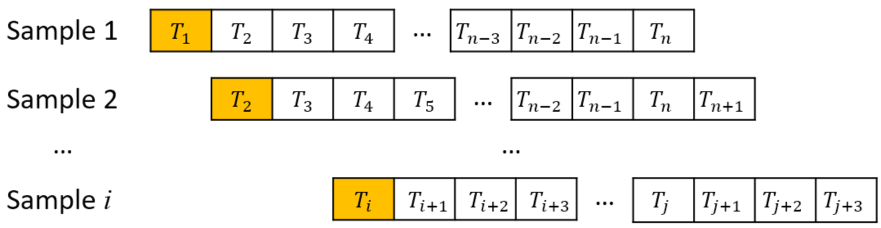

Figure 3.4: Dataset making process

As shown in Figure 3.4, after removing missing values, the training dataset is constructed using a sliding window approach. The window slides with a stride of one time unit to extract overlapping samples. In this study, the window size is set to 4,800, which corresponds to the predefined sample size. As a result, approximately 16,410 samples are generated for each data type in each building.

The dataset is then split into training and validation sets, with 20% of the samples allocated to the validation set—approximately 3,280 samples—while the remaining 80% are used for training.

Through these preprocessing steps, separate training and validation sets are created for each type of substation data. The analysis focuses on three key variables: Supply Water Temperature, Return Water Temperature, and Water Flow. These variables are collected from six buildings. To reduce redundancy and emphasize variation across building types, only one representative building is selected from each type, as the aim is to evaluate the model's effectiveness across different DHS-powered building configurations. A summary of the dataset configuration is presented in Table 3.1.

Table 3.1: Data summary

Building number	Building type	Total length	Total samples
1	Public	21209	16410
2	Public	21188	16389
3	Public	21164	16365
4	Public	21212	16413
7	Hotel	21189	16390
8	Office	21187	16388

In this thesis work, Building 1, 7, 8 are selected to represent public (*i.e.,* shop, gym, classroom, *etc.*), hotel and office types.

3.3 Experimental design

The experimental design is set for steps in the research process presented in Section 3.1, which includes not only the design methods utilized in this study, but also the implementation of models and clustering algorithms.

3.3.1 Test environment

The code version is Python 3.10.14 and it runs in the configuration of Intel i7-9750H @2.60GHz, 8GB RAM. The methods related to the model and clustering are in Tensorflow packages, including `Kmeans` package, `minisom` package, *etc*.

3.3.2 Prediction model

The prediction model used in this research is a SAE, consisting of two hidden layers in addition to the input and output layers. To balance model complexity and mitigate the risk of overfitting, a *L2* regularization term is applied to the input layer. A detailed summary of the SAE architecture is provided in Table 3.2.

Table 3.2: SAE architecture

Layer type	Neurons number	Parameter number [1]
Input layer	4800	864180
Hidden layer 1	180	3120
Latent layer	16	2896
Hidden layer 2	180	3060
Output layer	4800	868800

[1] The parameter number may change with different datasets.

After model construction, the training and validation datasets are fed into the SAE, and training is performed through iterative optimization. The MAE is selected as the loss function, and the loss is calculated for each individual sample.

A normality test on the loss values confirms that the distribution approximates a normal distribution. Therefore, the 3σ rule is adopted for anomaly thresholding instead of the IQR method. In this approach, a loss exceeding 1σ from the mean is considered indicative of a common anomaly, while a loss exceeding 2σ is classified as a significant anomaly. The identification of significant anomalies is directly based on this threshold, whereas the classification of common anomalies is further refined through clustering methods, due to their more diverse nature.

3.3.3 Clustering and interpretation

In this section, the clustering and interpretation processes after model prediction are demonstrated.

3.3.3.1 Clustering

After model prediction, a data frame is generated containing the anomaly flags alongside their corresponding samples. In this data frame, the MAE values serve as indicators of anomaly severity and are used as the basis for further analysis. To cluster the flagged anomalies according to their MAE values, a SOM is constructed. Figure 3.5 illustrates the SOM architecture used in this study. The SOM nodes are organized in a two-dimensional square grid. This structure facilitates the visualization and identification of distinct anomaly patterns within the dataset.

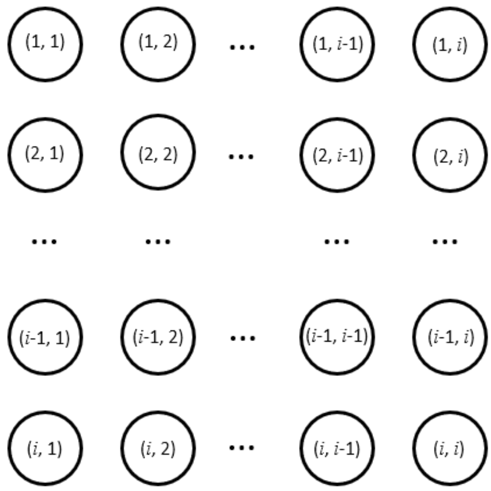

Figure 3.5: SOM used in the clustering

To get better result, the size of of SOM is calculated with the experience formula as

$$Size = 5\sqrt{N} \tag{3.1}$$

where N is the number of input. Once the SOM has been constructed, the MAE values of the anomaly samples are fed into the input layer for training. Through this process, the BMUs for all anomaly samples are obtained. These BMUs are subsequently clustered using the K-means algorithm to identify representative patterns of common anomalies.

The number of clusters, corresponding to the number of common operation deviations, is determined by analyzing the SOM distance map, which visually highlights the distribution and separation of the BMUs. Once the BMUs are grouped into the specified number of clusters, each cluster centroid serves as a representative of a distinct type of common anomaly.

Manual interpretation is then performed on the cluster centroids rather than on each individual anomaly sample. By assigning the interpretation results of the centroids to all samples within the same cluster, the overall workload for anomaly interpretation is significantly reduced, while still ensuring a meaningful classification of anomaly types.

3.3.3.2 Interpretation

As mentioned earlier, Anomaly interpretation in this study is conducted manually by comparing the time series of normal and anomalous samples. This approach is adopted due to the absence of labeled anomaly reports. In each evaluation time window, both the normal and anomalous time series are visualized within the same plot, allowing for direct comparison.

3.4 Evaluation framework

This section presents the evaluation for the substation data, anomaly detection results and the interpretation results. The evaluation results can show the data quality and the accuracy of the anomaly detection.

3.4.1 Substation data selection

The selection for substation data focuses on the missing values. For each dataset, the missing value rate is calculated as

$$MissingRate = 1 - \frac{N}{f_{sample} \cdot t} \tag{3.2}$$

where N means the total number of data points, f_{sample} means the sample rate and t means the sample time period. In general cases, the dataset which has a missing rate under 20% is regarded as the qualified dataset, and such dataset can be used for model training.

3.4.2 Results evaluation

The results evaluation in this study consists of two main components: anomaly detection performance and anomaly interpretation.

For the first evaluation, the focus is on assessing the model's performance. For each training dataset, a separate testing dataset is randomly sampled from the original data to validate the model. During training, the model's learning behavior is monitored through a training loss plot, as shown in Figure 3.6.

This plot allows early identification of potential issues such as overfitting or underfitting, thereby contributing to the assurance of model quality and generalization ability.

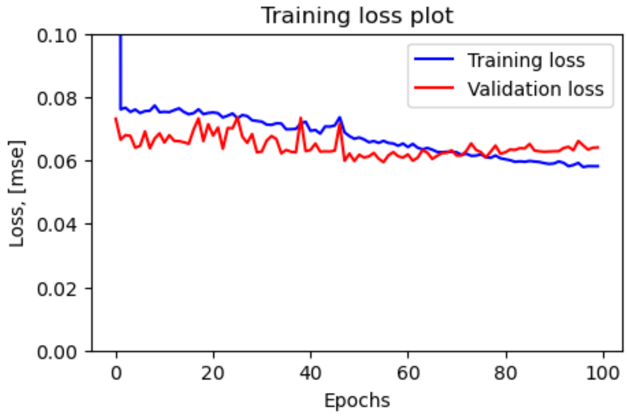

Figure 3.6: Training loss plot

For the second evaluation, the process relies primarily on expert knowledge and manual analysis. Through careful manual examination, anomaly types can be accurately identified and interpreted, minimizing the likelihood of significant errors. Additionally, prior studies, such as Sun *et al.* [14], provide standardized criteria for specific types of anomalies (see Table 2.2). These standards serve as reference metrics to validate the correctness and plausibility of the manual interpretations. Table 3.3 shows the metrics used to evaluate the interpretation, it contains four parameters: True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN). In this thesis, only anomaly samples marked by ADS are interpreted, therefore Precision (Calculated with Equation (3.3)) is selected as the metric to evaluate the results produced by ADS.

Table 3.3: Confusion Matrix

Reality / Prediction	Anomaly	Normal
Anomaly	TP	FP
Normal	FN	TN

$$Precision = \frac{TP}{TP + FP} \tag{3.3}$$

Chapter 4

Results and Analysis

In this chapter, we present the results and discuss them. The results include the simulation results of the research, and the analysis for reliability and validity.

4.1 Simulation results

In this section, the main simulation results of the proposed methodology are presented. The discussion begins with the detection of operational anomalies using unsupervised ML methods across three primary building types, and concludes with an analysis of potential causes that may have triggered these operational deviations. The period of this simulation is from January 2 to January 29, so the results mainly uncover the anomaly characters in winter.

4.1.1 Detection results

The detection results include both common and significant anomalies. Table 4.1 shows the model detection results for the three substation data types. In this table, the numbers of occurrences are shown for both common and significant deviations. The format "*x*S, *y*C" means there are *x* significant and *y* common anomalies occurred during the simulation period.

With the results of occurrence, it's also vital to identify the occurrence periods for significant deviations in order to have prompt maintenance. Table 4.2 shows the occurrence periods for all significant anomalies, where the start and end dates are presented. From the table, it's obvious that most of significant deviations occurred at the end of January, which may indicate the frequent changes of heating demand or the issues in the District Heating Network (DHN) of the three buildings.

Table 4.1: Anomaly detection results

Building	Detected anomaly		
	Water Flow	Supply Temperature	Return Temperature
Public	1S, 2C	1S, 2C	2C
Hotel	1S, 1C	1S, 2C	1S, 2C
Office	2C	1S, 2C	1S, 2C

Table 4.2: Significant anomaly occurred periods

Building	Significant deviation time (Unit: Day)		
	Water Flow	Supply Temperature	Return Temperature
Public	1.8 to 1.12	1.20 to 1.25	None
Hotel	1.20 to 1.26	1.20 to 1.26	1.21 to 1.26
Office	None	1.20 to 1.26	1.20 to 1.26

As shown in Figure 4.1, clustering based on the error between raw and filtered data for each DH energy metric enabled the identification of common operation deviations characteristic to each metric across different building categories. By analyzing the number of peaks in the density plot, human expertise can have a good preview in how many deviation types existed.

This crucial preliminary step facilitates human expertise by narrowing down the potential causes behind each deviation, thus effectively leveraging ML-based clustering to support more targeted and efficient interpretation.

After the preview of error density plot, the number of clusters for K-means can be defined accordingly. The clustering results are shown in Figure 4.2, in the format of SOM's distance map, where the bright color defines the boundary of clusters and the dark color marks the concentrations of samples.

With these results, the numbers of common and significant anomaly are determined, the interpretation then can be carried on these results in the next step.

4.1.2 Interpretation results

In this section, the interpretations of all detected anomalies are presented. A total of three anomaly categories are identified in Water Flow, three in

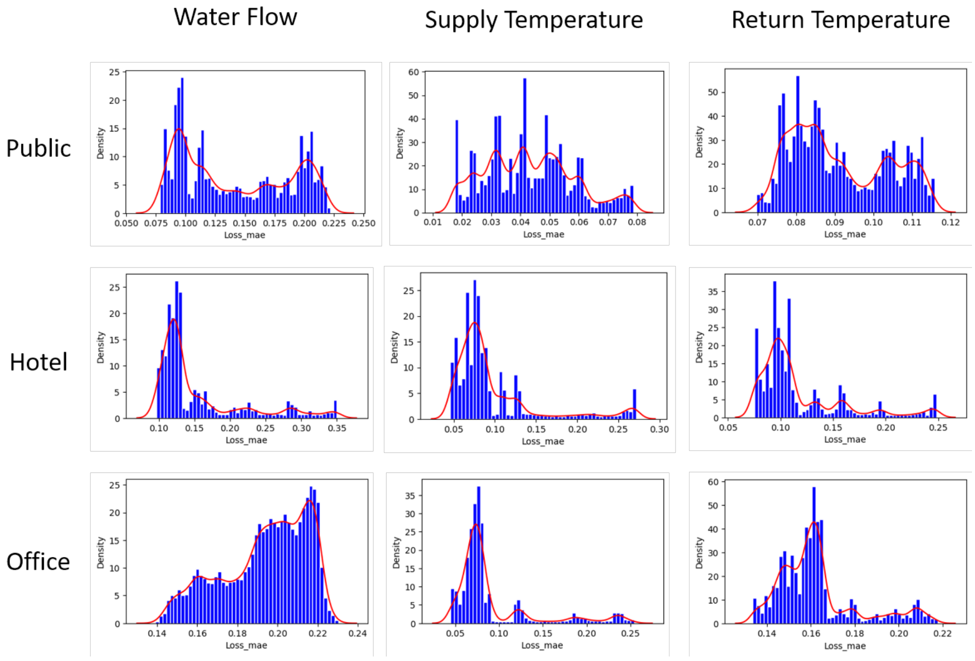

Figure 4.1: Density plots for common anomaly

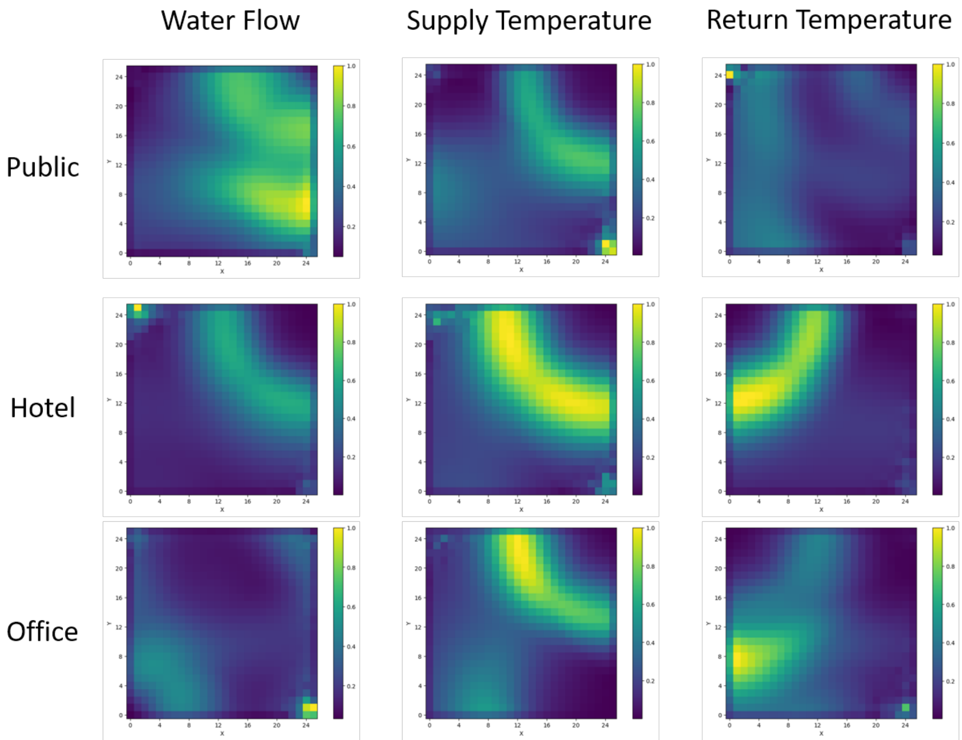

Figure 4.2: Error clustering results in January 2024

Supply Water Temperature, and another three in Return Water Temperature. It is important to note that significant operation deviation may share the same category as common deviations, as they can all belong to one of the existing clusters.

4.1.2.1 Water Flow

The Water Flow on the secondary side of the heat exchanger for each DHS-powered building energy consumer is primarily governed by the operation of the heating systems. Consequently, the data patterns for this parameter are mainly shaped by the heat demand profiles. However, additional factors, initially not considered, may also influence the Water Flow.

Figure 4.3 illustrates the Water Flow patterns of the three studied buildings over the month of January 2024, where “01006”, “07004” and “08005” is the numbers of Public, Hotel and Office, respectively. In the first and second subplots, significant fluctuations are observed, which the ADS was able to identify effectively. Such variations are typically attributed to poorly tuned Proportion Integration Differentiation (PID) controllers, commonly responsible for regulating Water Flow.

In addition, periods of prolonged steady-state flow have been detected as common operation deviation, often caused either by unusual heat load behaviors, such as continuously open windows, or by water leakage within the internal Heating, Ventilation and Air Conditioning (HVAC) loops. By contrast, the third subplot displays no notable or consistent operational deviations under the ADS configuration applied in this study.

Table 4.3: ADS performance in terms of Water Flow

	Public	Hotel	Office
Control Failure	*		
Leakage	*		
PID Control Issues	**	**, *	

Table 4.3 shows the interpretation result. In the result table, one star character (*) means common anomaly, two star characters (**) mean the significant anomaly. Noted that checked by the expert, there are no Water Flow operation deviation existed in Office.

Table 4.4 shows the measured precision for Water Flow. The Precision keeps high in Public and Hotel buildings, however the Office has a precision

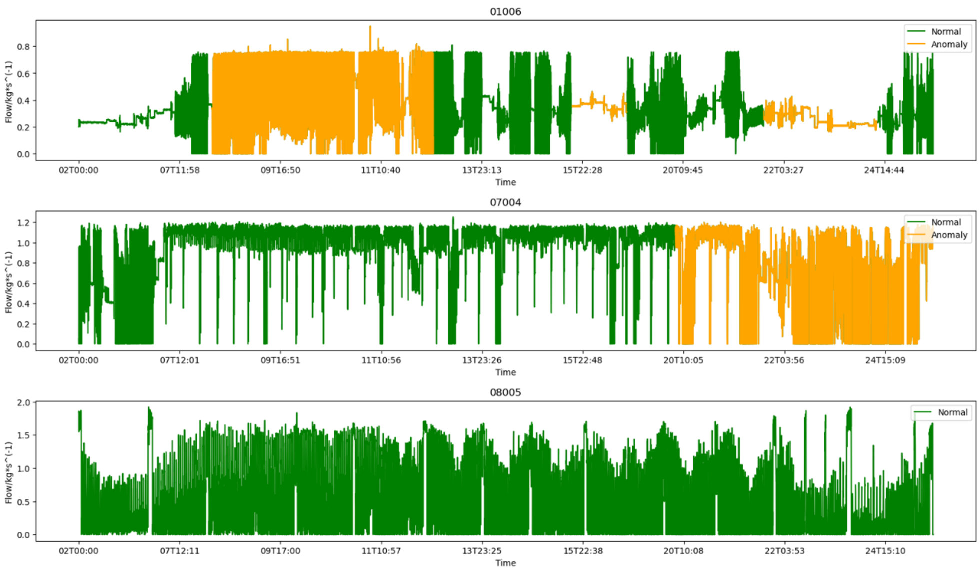

Figure 4.3: Water Flow detection plot

of 0 since the detected anomaly is confirmed as normal according to the expert.

Table 4.4: Precision in Water Flow

	Public	Hotel	Office
TP samples	702	1196	0
FP samples	33	54	10
Precision	95.51%	95.68%	0%

4.1.2.2 Supply Water Temperature

Ideally, the Water Supply Temperature on the secondary side of heat exchangers is predominantly determined by the Supply Temperature provided by the DHN. Nevertheless, other factors may interfere and influence the expected behavior of the data.

In addition to faulty sensors, excessively high Water Flow can lead to noticeable Supply Temperature drops. As a result, fluctuating flow rates also induce variations in the supply temperature. This relationship is directly influenced by the type and design specifications of the heat exchanger.

Consequently, any factor causing deviations in the Water Flow will also affect the supply temperature, with varying degrees depending on the building characteristics and the specific configurations of the DH substations. It is important to note that persistent supply temperature deviations could be attributed to heat losses within the DH supply network connecting different buildings. However, this study intentionally excludes any evaluation of DH network performance, as the primary focus is limited to the building level.

Table 4.5: ADS performance in terms of Supply Water Temperature

	Public	Hotel	Office
Control Instability	**, *		*
Abnormal Power Demand Pattern		**, *	**, *
Faulty Sensors	*	*	

Table 4.5 shows the interpretation result. In the result table, one star character (*) means common anomaly, two star characters (**) mean the significant anomaly. Figure 4.4 shows the detection results in three buildings: “01006”, “07004” and “08005” are tagged for Public, Hotel and Office. All three buildings are detected with significant anomalies after the model prediction and manual interpretation.

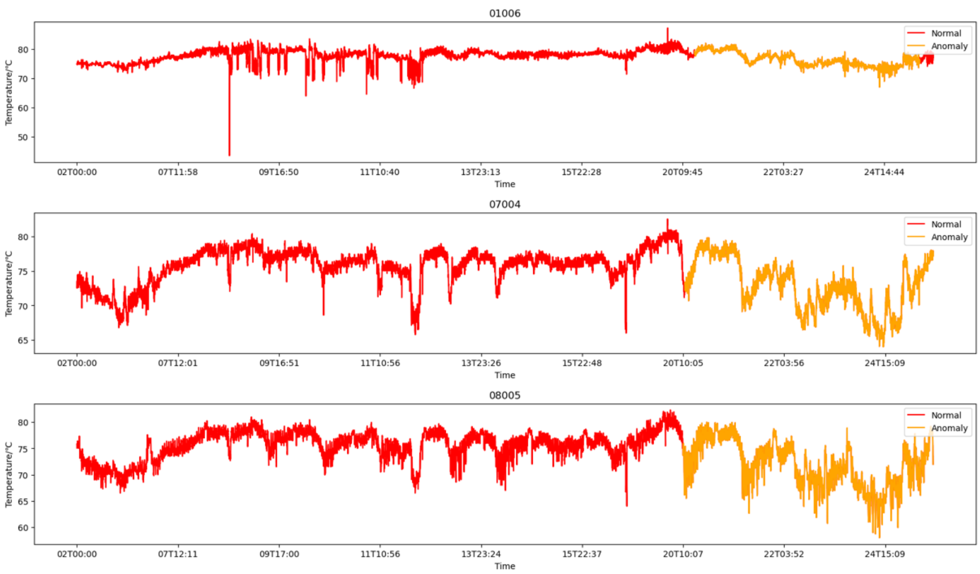

Figure 4.4: Supply Water Temperature detection plot

Table 4.6 shows the measured precision for Supply Water Temperature. Cause the Supply Water Temperature has less characters in change, the model managed to get high precision in detection results.

Table 4.6: Precision in Supply Water Temperature

	Public	Hotel	Office
TP samples	503	1821	1253
FP samples	12	423	41
Precision	97.67%	81.15%	96.83%

4.1.2.3 Return Water Temperature

Similar to the Water Flow behavior, return temperature data can also reflect performance deviations related to variations in heat demand: lower return temperatures typically correspond to higher heat demands, and vice versa. Therefore, factors contributing to Water Flow anomalies are often equally applicable to return temperature deviations.

However, an unusually high return temperature under normal heat demand conditions — particularly when the temperature difference (ΔT) between supply and return is small — can indicate unintended activation of a faulty bypass valve on the secondary side of the heat exchanger.

Another possible cause could be the presence of an alternative heat source independent of the DHS, which reduces the building's reliance on DH energy and subsequently narrows the ΔT.

Table 4.7: ADS performance in terms of Return Water Temperature

	Public	Hotel	Office
User Consumption	*	**	*
Foul Heat Exchanger	*	*	**, *
Activation of Bypass Valve		*	

Table 4.7 shows the interpretation result. In the result table, one star character (*) means common anomaly, two star characters (**) mean the significant anomaly. Figure 4.5 shows the detection results in three buildings: "01006", "07004" and "08005" are tagged for Public, Hotel and Office. The Public building has no significant anomaly detected whereas Hotel and Office have operation deviations detected.

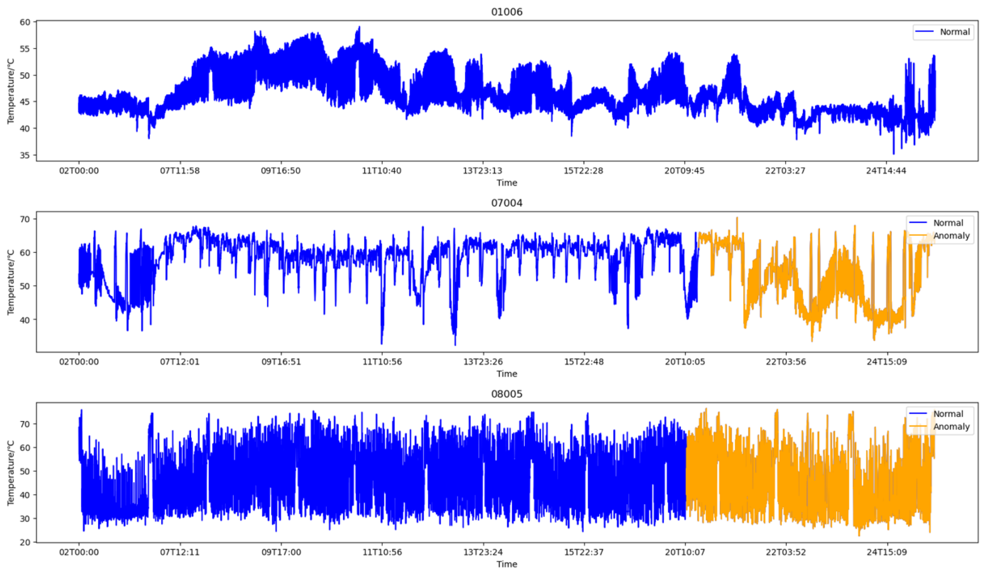

Figure 4.5: Return Water Temperature detection plot

Table 4.8 shows the measured precision for Return Water Temperature. Since Return Water Temperature is affected by various factors like Water Flow and user consumption change, the precision for Return Water Temperature detection model is medium without any extreme high or low values, and the values also various between Office and the other two buildings.

Table 4.8: Precision in Return Water Temperature

	Public	Hotel	Office
TP samples	487	2034	1431
FP samples	276	852	102
Precision	63.83%	70.48%	93.35%

4.2 Test–retest Reliability Analysis

In this thesis work, during the repeating test for the SAE models, the result stays nearly constant with the one described in the previous sections. The Test-Retest reliability values (Calculated with Pearson correlation) is shown in Table 4.9.

Table 4.9: The Test-Retest reliability values

	Water Flow	Supply Temperature	Return Temperature
Public	0.714	0.917	0.881
Hotel	0.867	0.903	1.000
Office	1.000	1.000	0.952

From the result in Table 4.9, the model prediction for the substation data has high correlation values in three variables, which means good reliability in retest. This ensures the good reliability in clustering and interpretation cause these two works are all based on the prediction results.

4.3 Validity Analysis

The validity of this thesis work is ensured through both model integrity and expert-driven manual analysis. During the training phase, the model's health is continuously monitored by examining the loss plots. This allows for early detection and correction of potential issues such as overfitting or underfitting, thereby maintaining the robustness of the predictive model. Moreover, the data in use cases are generated from real DH substations, which also ensures the reality and practicality of this study.

Furthermore, manual analysis provides an additional layer of validation by leveraging expert knowledge in DHS data interpretation and energy performance assessment. This dual approach—combining quantitative model evaluation with qualitative expert insight—ensures the reliability and credibility of the anomaly detection and interpretation results.

4.4 Discussion

The results demonstrate that the objectives of anomaly detection and interpretation across different building types have been successfully achieved through the application of SAE models and clustering techniques. The anomaly detection outcomes indicate that the majority of anomalies occurred during the period from January 20 to January 26, 2024.

This concentrated occurrence suggests that the substations of the studied buildings may have experienced operational issues during this time, potentially due to disturbances in the surrounding DHN. It is therefore recommended to further investigate the DHN at the community level in order to identify possible malfunctioning nodes or systemic irregularities that may have contributed to the observed anomalies.

Chapter 5

Conclusions and Future work

In this chapter, the conclusions of this thesis work are described, as well as the future work to optimize and better apply the ADS.

5.1 Conclusions

In this thesis work, the ADS contained two subsystems—one for anomaly prediction and one for anomaly clustering—were successfully developed and applied to real-world DHSes. Using substation time series data as input, these systems were able to accurately identify the timing and categorize the types of both common and significant anomalies. The results were further validated and interpreted with the aid of expert knowledge.

In conclusion, the proposed anomaly detection framework has demonstrated strong capability in identifying a wide range of anomalies in real substation data. The systems also exhibit high reliability and accuracy, making them practical for deployment in buildings and for supporting the routine monitoring and maintenance of DHS infrastructure.

5.2 Limitations

At present, the ADS operates automatically in the prediction and clustering stages. However, the interpretation phase still relies on manual analysis. This manual dependency significantly limits scalability and reduces efficiency, especially when processing large volumes of data.

In addition, the implemented systems are currently designed to analyze individual DHS in isolation. They are not yet capable of handling multiple interconnected systems within a broader DHN. This restricts their applicability for comprehensive network-level analysis and limits their potential for identifying systemic issues across multiple substations.

5.3 Future work

Due to the limitations in data and resource, many goals are unable to be achieved in this thesis work. In these section we will focus on some of the goals that can be achieved in future work.

5.3.1 All automatic systems

Due to the limited availability of literature and labeled datasets concerning specific types of anomalies and their behavioral patterns, the current systems are not yet capable of being trained with labeled anomaly data, nor can they automatically classify detected anomalies.

In future work, it is recommended that DHS operators and users collect and document more anomaly reports to build a labeled dataset, which would enable the systems to learn and classify new anomaly types more effectively. For automatic classification, models such as Support Vector Machine (SVM) and RF can be employed to improve accuracy. Incorporating these models would not only enhance the efficiency of anomaly interpretation but also improve the system's scalability and robustness when processing large-scale datasets.

5.3.2 DHN anomaly detection

Currently, the developed systems are capable of detecting anomalies within individual DHS units. However, if access to substation data from all end-users within a DHN becomes available, it would be highly valuable to extend the anomaly detection framework to cover the entire DHN. This expansion would enable the identification of problematic nodes within the network, allowing for timely maintenance and repair. Such proactive management would not only extend the operational lifespan of the DHN but also improve heating quality and reliability for all end-users within the community.

To realize this objective, comparisons among end-users are necessary. Implementing similarity metrics to evaluate behavioral patterns across substations would be an effective approach for enabling network-wide anomaly detection.

5.3.3 Optimization in accuracy

Regarding the ADS itself, there is still room for improving detection accuracy by considering practical aspects of DHS operation. For example, regular maintenance activities may cause temporary deviations in substation data that differ from typical “healthy” patterns, yet should not be classified as anomalies. Similarly, fluctuations in Return Water Temperature are often influenced by end-user consumption behaviors, which must be differentiated from actual system faults.

To address these issues, it is essential to incorporate knowledge of human-driven behaviors into the training process. By learning from patterns associated with user activities and operational procedures, the system can better distinguish between true anomalies and expected variations, thereby reducing false positives and improving overall detection reliability.

5.4 Reflections

This thesis primarily contributes to enhancing energy efficiency within DHS. According to statistical data on heating energy, a substantial portion of energy loss is attributable to undetected anomalies in DHS operations. The implementation of effective anomaly detection and prediction enables timely shutdowns and targeted maintenance, thereby reducing unnecessary energy consumption and ensuring consistent thermal comfort for users.

In addition, the system extends the operational lifespan of DHS infrastructure by minimizing the occurrence and duration of long-term anomalies. This aligns with the objectives of the Sustainable Development Goal (SDG), particularly those related to affordable and clean energy, sustainable cities, and responsible consumption. Furthermore, by facilitating faster and more efficient maintenance processes, the ADS reduces the need for manual labor and lowers maintenance costs, contributing to both economic and environmental sustainability.

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Appendix A

Supporting materials

The BibTeX references used in this thesis are attached.

Some source code relevant to this project can be found at https://github.com/yyttzzz/Master-Thesis in Jupyter Notebook format. The dataset used for this thesis can also be found via the same link.